

AidBridge — Database Design

Engine: PostgreSQL 15+ · **Extension:** uuid-oss
Design principle: 3NF cho tất cả bảng transactional; denorm có chủ ý cho read-heavy aggregates; query đơn giản và nhanh là ưu tiên hàng đầu.

1. Schema Overview

23 bảng · 10 kiểu ENUM · ~38 index

Group	Bảng
Auth	users , refresh_tokens
Profiles	volunteer_profiles , sponsor_profiles
Infrastructure	hubs , hub_staff , shelters , system_config
Catalog & Inventory	item_categories , hub_accepted_categories , hub_inventories , inventory_logs
Requests	sos_requests , sos_request_details *, aid_requests , aid_request_items
Donations	donations , donation_items
Missions	missions , dispatch_attempts
Communication	chat_messages , ratings , notifications

* sos_request_details — bảng mới (v3.1): vertical split từ sos_requests , chứa các cột TEXT ít truy vấn.

2. ENUM Types

ENUM	Giá trị
user_role	VICTIM , VOLUNTEER , SPONSOR , STAFF , ADMIN

ENUM	Giá trị
hub_status	ACTIVE , INACTIVE , EMERGENCY
urgency_level	CRITICAL , HIGH , MEDIUM , LOW
sos_status	PENDING , DISPATCHING , ASSIGNED , IN_PROGRESS , COMPLETED , CANCELLED
aid_status	PENDING , DISPATCHING , ASSIGNED , PICKED_UP , IN_TRANSIT , COMPLETED , CANCELLED
donation_status	REGISTERED , QR_GENERATED , RECEIVED , REJECTED
mission_type	RESCUE , DELIVERY
mission_status	PENDING , DISPATCHING , ASSIGNED , PICKING_UP , PICKED_UP , IN_TRANSIT , COMPLETED , CANCELLED
dispatch_response	PENDING , ACCEPTED , REJECTED , TIMEOUT
badge_level	BRONZE , SILVER , GOLD , PLATINUM

3. Table Definitions

GROUP 1 — Auth & Users

users

Column	Type	Constraint	Ghi chú
id	UUID	PK	DEFAULT gen_random_uuid()
full_name	VARCHAR(100)	NOT NULL	
email	VARCHAR(255)	UNIQUE	
phone_number	VARCHAR(20)	UNIQUE	
password_hash	VARCHAR(255)	NOT NULL	bcrypt — không log plain text
role	user_role	NOT NULL	

Column	Type	Constraint	Ghi chú
is_verified	BOOLEAN	DEFAULT FALSE	
is_active	BOOLEAN	DEFAULT TRUE	FALSE = bị Admin khóa
fcm_token	VARCHAR(500)		cập nhật mỗi lần login
avatar_url	VARCHAR(500)		
otp_code	VARCHAR(6)		mã OTP đang active
otp_type	VARCHAR(20)		EMAIL_VERIFY / PHONE_VERIFY / PASSWORD_RESET
otp_expires_at	TIMESTAMP		
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Constraints:

CHECK (email IS NOT NULL OR phone_number IS NOT NULL)

Tại sao gộp OTP vào users : OTP là single-use, short-TTL (10 phút), chỉ 1 OTP active per user tại một thời điểm. Gộp vào loại bỏ JOIN khi verify. Redis mirror OTP cho O(1) lookup; cột này là durable fallback.

refresh_tokens

Column	Type	Constraint
id	UUID	PK
user_id	UUID	FK(users) ON DELETE CASCADE
token_hash	VARCHAR(64)	UNIQUE NOT NULL

Column	Type	Constraint
device_info	TEXT	
expires_at	TIMESTAMP	NOT NULL
is_revoked	BOOLEAN	DEFAULT FALSE
created_at	TIMESTAMP	DEFAULT NOW()

Chỉ lưu SHA-256 hash — không bao giờ lưu raw token. Redis blacklist cho revocation tức thì; bảng này là durable backup khi Redis restart.

GROUP 2 — Role Profiles

volunteer_profiles

Column	Type	Constraint	Ghi chú
id	UUID	PK	
user_id	UUID	FK(users) UNIQUE	1:1 với users
is_online	BOOLEAN	DEFAULT FALSE	dispatch gate — chỉ query khi TRUE
current_lat	DECIMAL(9,6)		GPS mới nhất
current_lng	DECIMAL(9,6)		GPS mới nhất
vehicle_type	VARCHAR(50)		
total_tasks_completed	INT	DEFAULT 0	T-factor trong Priority Score
avg_rating	DECIMAL(3,2)	DEFAULT 0.00	R-factor (0–5)
avg_response_seconds	INT	DEFAULT 0	A-factor
created_at	TIMESTAMP	DEFAULT NOW()	

Column	Type	Constraint	Ghi chú
updated_at	TIMESTAMP	DEFAULT NOW()	

Dispatch Priority Score:

$S = (D \times 40\%) + (R \times 20\%) + (T \times 15\%) + (A \times 15\%) + (E \times 10\%)$

D = khoảng cách Haversine từ lat/lng | R = avg_rating | T = total_tasks_completed (normalized) | A = avg_response_seconds (inverse) | E = 0 (MVP đơn giản hóa)

sponsor_profiles

Column	Type	Constraint
id	UUID	PK
user_id	UUID	FK(users) UNIQUE
total_points	INT	DEFAULT 0
total_items_donated	INT	DEFAULT 0
donation_count	INT	DEFAULT 0
badge_level	badge_level	DEFAULT 'BRONZE'
created_at	TIMESTAMP	DEFAULT NOW()
updated_at	TIMESTAMP	DEFAULT NOW()

Các cột aggregate cập nhật atomic cùng transaction khi donation → RECEIVED .
Badge thresholds: BRONZE ≥ 1, SILVER ≥ 10, GOLD ≥ 25, PLATINUM ≥ 50 (donation_count).

GROUP 3 — Infrastructure

hubs

Column	Type	Constraint
id	UUID	PK
name	VARCHAR(100)	NOT NULL
address	TEXT	NOT NULL
lat	DECIMAL(9,6)	NOT NULL
lng	DECIMAL(9,6)	NOT NULL
status	hub_status	DEFAULT 'ACTIVE'
contact_phone	VARCHAR(20)	
created_at	TIMESTAMP	DEFAULT NOW()
updated_at	TIMESTAMP	DEFAULT NOW()

hub_staff

Column	Type	Constraint
id	UUID	PK
hub_id	UUID	FK(hubs)
user_id	UUID	FK(users)
is_available	BOOLEAN	DEFAULT TRUE
assigned_at	TIMESTAMP	DEFAULT NOW()
unassigned_at	TIMESTAMP	NULL = đang trực

Constraints:

```
-- Partial UNIQUE: mỗi người chỉ active ở 1 hub tại 1 thời điểm
CREATE UNIQUE INDEX ON hub_staff(hub_id, user_id) WHERE unassigned_at IS NULL;

-- Chỉ STAFF được assign
CHECK (EXISTS(SELECT 1 FROM users WHERE id = user_id AND role = 'STAFF'))
```

shelters

Column	Type	Constraint
id	UUID	PK
name	VARCHAR(100)	NOT NULL
address	TEXT	
lat	DECIMAL(9,6)	
lng	DECIMAL(9,6)	
current_capacity	INT	DEFAULT 0
max_capacity	INT	NOT NULL
has_electricity	BOOLEAN	DEFAULT FALSE
has_clean_water	BOOLEAN	DEFAULT FALSE
status	VARCHAR(15)	AVAILABLE / FULL / UNAVAILABLE
created_at	TIMESTAMP	DEFAULT NOW()
updated_at	TIMESTAMP	DEFAULT NOW()

Constraint: CHECK (current_capacity <= max_capacity)

system_config

Column	Type	Constraint
key	VARCHAR(100)	PK

Column	Type	Constraint
value	TEXT	NOT NULL
description	TEXT	
updated_at	TIMESTAMP	DEFAULT NOW()

Pre-seeded keys:

Key	Default
dispatch.weight.distance	0.40
dispatch.weight.rating	0.20
dispatch.weight.tasks	0.15
dispatch.weight.response_time	0.15
dispatch.radius.step1_km ... step4_km	1/3/5/10
dispatch.sos.window_seconds	30
dispatch.delivery.window_1	15
dispatch.delivery.window_batch	20
otp.ttl_minutes	10

GROUP 4 — Catalog & Inventory

item_categories

Column	Type	Constraint
id	UUID	PK
parent_id	UUID	FK(self) nullable
name	VARCHAR(100)	NOT NULL
name_vi	VARCHAR(100)	NOT NULL

Column	Type	Constraint
unit	VARCHAR(20)	NOT NULL
is_leaf	BOOLEAN	DEFAULT TRUE
created_at	TIMESTAMP	DEFAULT NOW()

Cây 2 cấp:

Thức ăn (is_leaf=FALSE)

- └─ Gạo (is_leaf=TRUE, unit='Kg')
- └─ Mì tôm (is_leaf=TRUE, unit='Thùng')
- └─ Đồ hộp (is_leaf=TRUE, unit='Hộp')

Trigger fn_mark_parent_non_leaf : khi INSERT child → tự SET parent.is_leaf = FALSE .
Chỉ leaf categories được dùng trong hub_inventories , aid_request_items , donation_items .

hub_accepted_categories

Column	Type	Constraint
hub_id	UUID	FK(hubs)
item_category_id	UUID	FK(item_categories)

PK: (hub_id, item_category_id) — composite.

Admin cấu hình per-hub: trạm nào nhận loại hàng nào. Bắt buộc cho Smart Hub Selection query.

hub_inventories

Column	Type	Constraint
id	UUID	PK
hub_id	UUID	FK(hubs)
item_category_id	UUID	FK(item_categories)

Column	Type	Constraint
current_quantity	INT	DEFAULT 0
low_stock_threshold	INT	DEFAULT 0
updated_at	TIMESTAMP	DEFAULT NOW()

Constraints:

```
UNIQUE (hub_id, item_category_id)
CHECK (current_quantity >= 0)
```

inventory_logs

Column	Type	Constraint	Ghi chú
id	UUID	PK	
hub_inventory_id	UUID	FK(hub_inventories)	
change_type	VARCHAR(10)	NOT NULL	INBOUND / OUTBOUND
quantity_delta	INT	CHECK > 0	luôn dương; hướng xác định bởi change_type
reference_type	VARCHAR(10)		DONATION / MISSION
reference_id	UUID		FK tới donation hoặc mission
performed_by	UUID	FK(users)	Staff thực hiện
quantity_after	INT	NOT NULL	snapshot sau transaction
notes	TEXT		
created_at	TIMESTAMP	DEFAULT NOW()	Append-only

quantity_after là denorm có chủ ý: query tồn kho tại bất kỳ thời điểm nào = O(1) thay vì O(n) replay.

GROUP 5 — Requests

sos_requests — HOT TABLE (v3.1: Vertical Split)

v3.1 thay đổi: Các cột TEXT ít dùng đã được tách sang `sos_request_details` . Bảng này chỉ giữ các cột truy vấn nóng phục vụ dispatch algorithm và heatmap.

Column	Type	Constraint	Ghi chú
id	UUID	PK	DEFAULT gen_random_uuid()
requester_id	UUID	FK(users) nullable	NULL = Guest
victim_lat	DECIMAL(9,6)	NOT NULL	hot: dispatch origin
victim_lng	DECIMAL(9,6)	NOT NULL	hot: dispatch origin
people_count	INT	CHECK > 0	hot: urgency weighting
is_on_behalf	BOOLEAN	DEFAULT FALSE	
urgency_level	urgency_level		AI-classified; hot: Priority Score
status	sos_status	DEFAULT 'PENDING'	hot: dispatch & heatmap filter
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Constraint: CHECK (people_count > 0)

sos_request_details — COLD TABLE (v3.1: mới)

Mục đích: Lưu các cột TEXT lớn, ít dùng — chỉ JOIN khi cần hiển thị chi tiết (Volunteer mission screen, Admin dashboard). Quan hệ 1:1 bắt buộc với `sos_requests` .

Column	Type	Constraint	Ghi chú
sos_request_id	UUID	PK · FK(sos_requests)	1:1 mandatory
requester_name	VARCHAR(100)	NOT NULL	
requester_phone	VARCHAR(20)	NOT NULL	

Column	Type	Constraint	Ghi chú
victim_name	VARCHAR(100)		khi is_on_behalf=TRUE
victim_phone	VARCHAR(20)		
victim_address	TEXT		
description	TEXT	NOT NULL	raw text từ Victim
ai_summary	TEXT		AI-generated tóm tắt
image_url	VARCHAR(500)		ảnh đính kèm (optional)

aid_requests

Column	Type	Constraint
id	UUID	PK
requester_id	UUID	FK(users) NOT NULL
lat	DECIMAL(9,6)	
lng	DECIMAL(9,6)	
address	TEXT	
adults_count	INT	DEFAULT 0
elderly_count	INT	DEFAULT 0
children_count	INT	DEFAULT 0
notes	TEXT	
urgency_level	urgency_level	
status	aid_status	DEFAULT 'PENDING'
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

Constraint: CHECK (adults_count + elderly_count + children_count > 0)

aid_request_items

Column	Type	Constraint
id	UUID	PK
aid_request_id	UUID	FK(aid_requests) ON DELETE CASCADE
item_category_id	UUID	FK(item_categories)
quantity	INT	CHECK > 0
created_at	TIMESTAMP	

GROUP 6 — Donations

donations

Column	Type	Constraint	Ghi chú
id	UUID	PK	
sponsor_id	UUID	FK(users)	
hub_id	UUID	FK(hubs)	
hub_name	VARCHAR(100)		DENORM — snapshot khi QR_GENERATED
estimated_delivery_at	DATE		
qr_code_token	VARCHAR(255)	UNIQUE	
status	donation_status	DEFAULT 'REGISTERED'	
received_by	UUID	FK(users) nullable	
received_at	TIMESTAMP		
rejection_reason	TEXT		
created_at	TIMESTAMP		

Column	Type	Constraint	Ghi chú
updated_at	TIMESTAMP		

hub_name
: Snapshot tên Hub tại thời điểm Sponsor chọn Hub (QR_GENERATED). QR Screen (§6.3) dùng trực tiếp mà không JOIN hubs .

Lifecycle:
REGISTERED
→
QR_GENERATED
→
RECEIVED
(kho cộng)
|
REJECTED
(lý do ghi log)

donation_items

Column	Type	Constraint
id	UUID	PK
donation_id	UUID	FK(donations) ON DELETE CASCADE
item_category_id	UUID	FK(item_categories)
quantity	INT	CHECK > 0
expiry_date	DATE	
condition_notes	TEXT	
image_url	VARCHAR(500)	
created_at	TIMESTAMP	

GROUP 7 — Missions

missions

Column	Type	Constraint	Ghi chú
id	UUID	PK	
mission_type	mission_type	NOT NULL	RESCUE / DELIVERY
sos_request_id	UUID	FK(sos_requests) nullable	RESCUE only

Column	Type	Constraint	Ghi chú
aid_request_id	UUID	FK(aid_requests) nullable	DELIVERY only
volunteer_id	UUID	FK(volunteer_profiles)	
hub_id	UUID	FK(hubs) nullable	DELIVERY only
status	mission_status	DEFAULT 'PENDING'	
qr_code_token	VARCHAR(255)	UNIQUE	single-use
priority_score	DECIMAL(8,4)		snapshot tại thời điểm assign
snapshot_lat	DECIMAL(9,6)		DENORM — tọa độ đích tại accepted_at
snapshot_lng	DECIMAL(9,6)		DENORM
snapshot_address	TEXT		DENORM — địa chỉ đích để hiển thị
snapshot_requester_name	VARCHAR(100)		DENORM — tên nạn nhân/người gửi
snapshot_requester_phone	VARCHAR(20)		DENORM — SĐT liên lạc
accepted_at	TIMESTAMP		
picked_up_at	TIMESTAMP		DELIVERY only
completed_at	TIMESTAMP		
cancelled_at	TIMESTAMP		
cancellation_reason	TEXT		
confirmation_image_url	VARCHAR(500)		ảnh xác nhận giao hàng
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Snapshot columns (DENORM): Được populate **tại thời điểm** Volunteer accept (accepted_at).
Phục vụ Live Tracking (§4.4) và Mission Screen (§5.3) poll liên tục mà không JOIN sang
sos_requests / aid_requests .

CHECK constraints:

```
-- RESCUE: không có hub, không có AID request
-- DELIVERY: bắt buộc có hub và AID request
CHECK (
  (mission_type = 'RESCUE'   AND sos_request_id IS NOT NULL AND aid_request_id IS NULL AND hub_
OR
  (mission_type = 'DELIVERY' AND aid_request_id IS NOT NULL AND sos_request_id IS NULL AND hub_
)
```

dispatch_attempts

Column	Type	Constraint
id	UUID	PK
mission_id	UUID	FK(missions) ON DELETE CASCADE
volunteer_id	UUID	FK(volunteer_profiles)
dispatch_type	VARCHAR(15)	BROADCAST / SEQUENTIAL
batch_number	INT	DEFAULT 1
radius_km	DECIMAL(5,2)	
priority_score	DECIMAL(8,4)	snapshot
sent_at	TIMESTAMP	DEFAULT NOW()
response	dispatch_response	DEFAULT 'PENDING'
responded_at	TIMESTAMP	

Full audit log cho mỗi volunteer được notify. Admin dùng để phân tích hiệu quả dispatch.

GROUP 8 — Communication

chat_messages

Column	Type	Constraint
id	UUID	PK
mission_id	UUID	FK(missions) ON DELETE CASCADE
sender_id	UUID	FK(users)
message_type	VARCHAR(5)	TEXT / IMAGE
message_text	TEXT	
image_url	VARCHAR(500)	
is_read	BOOLEAN	DEFAULT FALSE
created_at	TIMESTAMP	

CHECK:

```
CHECK (  
  (message_type = 'TEXT' AND message_text IS NOT NULL AND image_url IS NULL)  
  OR  
  (message_type = 'IMAGE' AND image_url IS NOT NULL AND message_text IS NULL)  
)
```

ratings

Column	Type	Constraint
id	UUID	PK
mission_id	UUID	FK(missions) UNIQUE
rater_id	UUID	FK(users)
ratee_id	UUID	FK(volunteer_profiles)
score	INT	CHECK (score BETWEEN 1 AND 5)

Column	Type	Constraint
comment	TEXT	
created_at	TIMESTAMP	

UNIQUE(mission_id) đảm bảo mỗi mission chỉ được đánh giá 1 lần.

notifications

Column	Type	Constraint
id	UUID	PK
user_id	UUID	FK(users) ON DELETE CASCADE
title	VARCHAR(200)	NOT NULL
body	TEXT	
related_type	VARCHAR(20)	MISSION / SOS / DONATION / AID_REQUEST
related_id	UUID	
is_read	BOOLEAN	DEFAULT FALSE
created_at	TIMESTAMP	

4. Indexing Strategy

Dispatch (query nóng nhất)

Index	Column	Loại	Mục đích
idx_vol_online	volunteer_profiles(current_lat, current_lng) WHERE is_online=TRUE	PARTIAL BTREE	is_online filter + bounding box

Index	Column	Loại	Mục đích
			distance pre-filter
idx_sos_location	sos_requests(victim_lat, victim_lng, status) WHERE status IN ('PENDING','DISPATCHING')	PARTIAL BTREE	Heatmap + dispatch origin (hot table sau split)
idx_aid_location	aid_requests(lat, lng, status) WHERE status IN ('PENDING','DISPATCHING')	PARTIAL BTREE	Heatmap + dispatch origin
idx_hubs_location	hubs(lat, lng) WHERE status='ACTIVE'	PARTIAL BTREE	Smart Hub Selection

Read (frequent lookups)

Index	Column
idx_missions_volunteer	missions(volunteer_id, status)
idx_missions_sos	missions(sos_request_id) WHERE NOT NULL
idx_missions_aid	missions(aid_request_id) WHERE NOT NULL
idx_dispatch_pending	dispatch_attempts(mission_id) WHERE response='PENDING'
idx_donations_sponsor	donations(sponsor_id, created_at DESC)
idx_donations_hub_status	donations(hub_id, status)
idx_inventory_hub	hub_inventories(hub_id)
idx_inv_low_stock	hub_inventories(hub_id, current_quantity) WHERE current_quantity <= low_stock_threshold
idx_inv_logs_ref	inventory_logs(reference_type, reference_id)
idx_chat_mission	chat_messages(mission_id, created_at ASC)
idx_notifications_user	notifications(user_id, created_at DESC) WHERE is_read=FALSE

5. Business Rules — DB Level

Rule	Cơ chế
Ít nhất email hoặc phone	CHECK (email IS NOT NULL OR phone_number IS NOT NULL)
Tồn kho không âm	CHECK (current_quantity >= 0) trên hub_inventories
RESCUE không có hub; DELIVERY bắt buộc có hub	CHECK 2 nhánh trên missions
Mission link đúng loại request	Cùng CHECK trên missions
Mỗi mission chỉ 1 rating	UNIQUE(mission_id) trên ratings
Score 1–5	CHECK (score BETWEEN 1 AND 5)
Tồn kho log delta > 0	CHECK (quantity_delta > 0)
Item qty > 0	CHECK (quantity > 0) trên donation_items , aid_request_items
SOS people_count > 0	CHECK (people_count > 0)
Shelter capacity	CHECK (current_capacity <= max_capacity)
1 active staff per hub per person	Partial UNIQUE index (hub_id, user_id) WHERE unassigned_at IS NULL
Chỉ STAFF vào hub_staff	CHECK role = 'STAFF'
Refresh token = hash only	Application-enforced
sos_request_details bắt buộc với mỗi SOS	Application-enforced: INSERT 2 bảng trong cùng transaction
missions.snapshot_* populated khi accept	Application-enforced: populate khi Volunteer accept, trước khi UPDATE status → ASSIGNED
updated_at auto-maintain	Trigger fn_set_updated_at trên mọi bảng mutable

6. Redis Layer (không trong PostgreSQL)

Dữ liệu	Cấu trúc Redis	TTL
JWT blacklist	SET revoked:{jti}	Đến khi token hết hạn
Volunteer live location	GEOADD volunteer_locations {lat} {lng} {vol_id}	Cho đến khi is_online=FALSE
Dispatch state	HASH dispatch:{mission_id}	Cho đến khi mission kết thúc
WebSocket pub/sub	Pub/Sub channels	Session lifetime
Dispatch Redis Lock	SET lock:mission:{id} NX EX 5	5 giây (anti race condition)